

MATHEMATICS
Introduction to Modern Algebra I
Midterm 3 Solutions.

1) Write the class equation and then use the class equation to show that if G is a group of order p^n and p is a prime number, then $Z(G)$ the center of G is nontrivial (i.e. $Z(G) \neq \{e\}$).

Solution:

$$|G| = \sum_a |G|/|C(a)| = |Z(G)| + \sum_{a \notin Z(G)} |G|/|C(a)|.$$

If $a \notin Z(G)$, then $C(a) \neq G$ and so p divides $|G|/|C(a)|$. Clearly p divides $|G|$. From the class equation, we have that p divides $|Z(G)|$. Since $e \in Z(G)$, we have $|Z(G)| > 0$ and hence $|Z(G)| \geq p$ so that $Z(G) \neq \{e\}$.

2) Let $\phi : G \rightarrow G'$ be a surjective homomorphism and N' a normal subgroup of G' . Show that $\phi^{-1}(N')$ is a normal subgroup of G .

Solution: Closed under multiplication: If $h, k \in \phi^{-1}(N')$ then $\phi(h), \phi(k) \in N'$ so that $\phi(hk) \in N'$ i.e. $hk \in \phi^{-1}(N')$.

Closed under inverses: If $h \in \phi^{-1}(N')$ then $\phi(h) \in N'$ so that $\phi(h^{-1}) = \phi(h)^{-1} \in N'$ and hence $h^{-1} \in \phi^{-1}(N')$.

If $n \in \phi^{-1}(N')$ then $\phi(n) \in N'$. Let $g \in G$, then $\phi(gng^{-1}) = \phi(g)\phi(n)\phi(g)^{-1} \in N'$ as N' is normal in G' . Thus $gng^{-1} \in \phi^{-1}(N')$. We have shown that $g\phi^{-1}(N')g^{-1} \subset \phi^{-1}(N')$ and hence $\phi^{-1}(N')$ is normal in G .

3) Let N be a subgroup of index 2 of a group G . Show that N is normal in G . Give an example of a non-normal subgroup of index 3.

Solution: If N has index 2 then it has exactly 2 left (resp. right) cosets. So $G = N + gN = N + Ng$ so that $gN = Ng$.

The subgroup of S_3 generated by $(1, 2)$ has index 3 but is not normal (eg. $(1, 3)(1, 2)(1, 3) = (2, 3) \notin \langle (1, 2) \rangle$).

4) Use the result of question 1) to show that if G is a group of order p^2 where p is a prime, then G is abelian.

Solution: We know that $|Z(G)| > 1$ and hence $|Z(G)| \in \{p, p^2\}$. If $|Z(G)| = p^2$, then G is abelian and we are done.

If $|Z(G)| = p$, then let $g \in G - Z(G)$. Since $C(g) \supset g \cup Z(G)$ the order of the subgroup $C(g)$ is at least $p + 1$ and hence it is p^2 . Thus $C(g) = G$ so that $g \in Z(G)$. This is the required contradiction.

5) Which one of the following is $\mathbb{Z}_{10} \times \mathbb{Z}_{100}$ isomorphic to? $\mathbb{Z}_{1000}$, $\mathbb{Z}_{20} \times \mathbb{Z}_{50}$, $\mathbb{Z}_{40} \times \mathbb{Z}_{25}$ or $\mathbb{Z}_2^3 \times \mathbb{Z}_5^2$. Explain your answer!

Solution:

$$\begin{aligned}\mathbb{Z}_{10} \times \mathbb{Z}_{100} &\cong \mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_4 \times \mathbb{Z}_{25} \cong \\ &\mathbb{Z}_4 \times \mathbb{Z}_5 \times \mathbb{Z}_2 \times \mathbb{Z}_{25} \cong \mathbb{Z}_{20} \times \mathbb{Z}_{50}.\end{aligned}$$

6) Define a surjective homomorphism $\phi : \mathbb{Z} \rightarrow \mathbb{Z}_2 \times \mathbb{Z}_3$ and prove that there is no surjective homomorphism $\phi : \mathbb{Z} \rightarrow \mathbb{Z}_2 \times \mathbb{Z}_2$.

Solution: $\mathbb{Z}_2 \times \mathbb{Z}_3 \cong \mathbb{Z}_6 \cong \mathbb{Z}/6\mathbb{Z}$. So let $\phi(x) = x + 6\mathbb{Z} \in \mathbb{Z}/6\mathbb{Z}$. (If you prefer, let $\phi(x) = (x + 2\mathbb{Z}, x + 3\mathbb{Z}) \in \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$.)

Since $\mathbb{Z}$ is cyclic, its image $\phi(\mathbb{Z}) \subset \mathbb{Z}_2 \times \mathbb{Z}_2$ must be cyclic. Since $\mathbb{Z}_2 \times \mathbb{Z}_2$ is not cyclic, ϕ can not be surjective.